

Team 1

Rubber Duck Programmers

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Exploring the Problem Space

1. Phishing/Dark Patterns

- a. The problem is that there are bad actors on the internet who are intelligent and skilled enough to deceive uninformed users and harm them socially, financially, or worse. The problem exists because it is increasingly easy through the tools at our disposal, such as Artificial Intelligence (AI), to create professional copies of legitimate websites where people feel safe to enter personal information into to instead receive that information themselves to commit crimes. There is nothing available that has done a successful job at educating those that are uninformed in the telltale signs and dangers of phishing and dark patterns.
- b. The people affected are mostly the elderly and the youth, though anyone without a technical understanding of the internet are affected as well. Most phishing schemes and dark patterns exploit how the average person does not understand how certain aspects of the internet work. This is especially true for elderly people with little experience with technology and young, more impressionable children. However, because each of these people is a father, grandfather, daughter, or aunt to someone, these problems bleed out to eventually affect even those not directly involved. One person falling victim to phishing will inevitably affect their entire family as well. Everyday consumers are also affected by the dark patterns used to psychologically manipulate them into acting in the way the business desires them to.
- c. Analog solutions include but are not limited to, tellers at banks are trained to spot potential phishing attempts that take the victim offline. There are signs at gift card stations in stores now that display warnings to purchasers that they may be a victim of a scam. There are built-in email and phishing filters to help prevent those emails from even being delivered and read. The FTC has targeted deceptive practices and subscription practices that make it difficult to cancel (Gym memberships). However, all these current “solutions” are limited in their ability to educate people on how they can prevent these scams themselves.
- d. There have been observations of people just clicking through dialog boxes not paying attention to what they have accepted just because they do not seem to care enough, similar to the stereotypical terms and conditions. In having some conversations, it seems that there is a growing trend of not being bothered too much by giving away privacy rights because of the notion that it is already too late to have anything be private anymore anyway. People also seem to be generally uneducated on the existence of dark patterns, some seem to know that they exist just through deductions but did not know the intentionality behind it.

2. Memory

- a. The problem is that as people age or if they are affected by a disease or disorder of some kind, they will lose their ability to remember events and tasks in the short term. Most people do not have the privilege to personally take care of these people as they are preoccupied with ensuring their own survival as well. The problem still exists because it is tough to solve an issue that a person does not even realize has occurred for them. The majority of the group most affected by this problem are also typically the ones that would have the most difficulty with adopting new technology made to help them.
- b. Primarily older people with declining mental strength due to either aging or diseases.
- c. There are many solutions in this space currently as it is a problem people have been dealing with since the beginning of time. Analog solutions include paper calendars and daily planners, whiteboards, checklists, sticky notes, medication charts and daily medication boxes, labels on drawers/cabinets, memory books, and so on. There is probably just as much if not more in the digital space such as calendar apps, to do lists, medication reminders, voice notes, location sharing, appointment reminders, notes app, family messages, voice assistants, etc.
- d. Often the people who face these issues require others to help them navigate tasks and problems they face because of cognition issues.

3. Procrastination

- a. The problem is that people are required to be more productive than they typically desire to. There are mental hurdles that come into play when a person decides that they want to take control of their life and turn their “wasted time” into “productive time”. People tend to have trouble either finding the motivation or figuring out where to start. This exists because we are not perfect machines, we are flawed humans and because of that we have obstacles that we put in front of ourselves, even if it’s subconsciously, that we need to overcome.
- b. Students and professionals, however everyone probably struggles with this problem even if it’s at a miniscule proportion. This also does not have to be so aggressive that it is even a problem, some people like to be ahead of the curve and be extra productive so this would help them as well.
- c. Analog solutions include to-do lists, calendars, planners, etc to help motivate you to check tasks off to give you a little sense of accomplishment and encouragement to continue, techniques like “just 5 minutes” where you try to trick yourself into working by breaking up a larger task into a small subtask hoping that it will cause you to snowball into doing more, accountability partners and public commitments

which are intended to help keep you true to your word when you say you will do something. Digital solutions include to do list, calendar, and planner apps and habit trackers that are intended to do the same thing as analog but on your electronic device. Screen-time controls and website blockers to help stop you from making poor decisions and bar you from any distractions.

- d. Through a conversation with a friend, he wanted to stop being such a procrastinator and wanted to start being more productive so he downloaded todoist and used that for a week before not having the motivation to want to set up his to do list anymore and fell out of the habit of caring to check the app daily. Finding a way to keep people motivated long enough to build it into a habit would be useful. There are numerous studies and books, including Atomic Habits, which propose that after 2 months then a task becomes a daily habit which would lower the barrier to entry for it.

Problem Overview

Description:

Many people across the world are struggling with memory and cognitive issues for various reasons, including medical conditions such as dementia and Alzheimer's disease. Although some changes in memory can occur with age, severe memory impairment is not a normal part of aging. These people will often forget to perform necessary tasks throughout their day, such as taking medicine, eating meals, or attending appointments. In extreme cases, these people may even become disoriented; they wander away from their homes with little means of being able to be found.

Our project focuses on individuals whose memory difficulties interfere with essential daily activities and the caregivers who support them. The problem involves not only remembering to perform a task but also remembering whether it has already been completed. This can make everyday routines difficult to manage independently.

Importance:

Severe memory issues can have drastic consequences for the person suffering from them. Memory-failing individuals could forget to eat, take medicine, lock doors, or even wander off and forget how to get back. These difficulties can affect both their safety and their ability to maintain independence.

This also places additional responsibilities on caregivers, who may need to provide reminders, assist with daily activities, and check whether important tasks have been completed. The level of support needed differs from person to person. Understanding both the individual's difficulties and the caregiver's responsibilities is therefore important when investigating this problem.

Existing Solutions:

Currently, there are several physical solutions to these issues. Examples include calendars, written reminders, checklists, and pill organizers. Calendars and notes make important information visible, but their usefulness depends partly on whether the individual notices and uses them. Placing reminders in familiar, frequently used locations may help them become part of an everyday routine.

Pill organizers separate medication into compartments, while automated dispensers can make scheduled medication available at times and provide an alarm. On the digital side, GPS tracking, smart-home devices, and digital reminders exist. These solutions provide different forms of

support: medication dispensers help organize scheduled doses, reminders' prompt activities, and GPS tracking offers a way to locate individuals who become lost. Some medication systems can also alert another person when medication has not been accessed, or the device needs attention.

However, these solutions still rely on the individual's memory to use them.

A written calendar must be checked, and a reminder does not by itself establish that the task was completed. This creates an important question for our research: how much responsibility does each approach place on the individual, and when does a caregiver need to intervene? We also need to investigate practical requirements, such as who sets up charges, carries, or maintains a device.

Furthermore, a study by Alex Hall and colleagues published in the *International Journal of Nursing Studies* found that monitoring technologies used in dementia care homes generated frequent alarms that placed a burden on staff. Although these technologies have the potential to support safety and freedom of movement, the study shows that their use can also create additional work. Staff relied on their knowledge of residents and circumstances to help manage the alarms. Hall et al., 2017

These limitations suggest that providing a reminder or monitoring device is only one part of the problem. Further research is needed to understand how individuals respond to existing aids, how task completion is confirmed, and how caregivers provide support when difficulties occur.

Stakeholders and Users

Potential Stakeholders:

- Primary Users: People with severe memory/cognitive impairment
 - o Individuals whose memory/cognitive function has declined due to age.
 - o Individuals whose memory/cognitive function has declined due to dementia or Alzheimer's
 - o Individuals who have a high likelihood of forgetfulness or becoming disoriented/lost.
- Secondary Users: Caregivers
 - o Family, friends, or professional caregivers of the primary user
 - o Individuals who would like to monitor the primary user to ensure they remember tasks
 - o Individuals who would like to track the primary user to ensure they do not wander away.
- Tertiary Users: Healthcare and care organizations
 - o Medical providers, assisted living organizations, and other healthcare organizations.
 - o Entities interested in the safety of clients and lowering the need for continuous monitoring by caregivers.

In 2026, Bouzida et al. researched how people with mild cognitive impairment and their care partners experienced frustration with technology and other systems. The research found that the participants became frustrated in areas involving communication, navigating changes in their abilities, and using technology. This research shows that when developing an app, both the impaired person's and the caregiver's needs must be considered in development. As such, we determined that our primary user would be the impaired person, since their needs would require the most consideration. Next, the caregiver would likely be a secondary user, since they wouldn't be reliant on the software, but would be utilizing it throughout the day as a means of checking on their loved one.

As for Healthcare organizations, implementation of technology for the sake of monitoring in care homes has been a developing topic for some time. In 2017, a study done by Alex Hall and others for the International Journal of Nursing Studies showed that the implementation of monitoring technologies in care homes for dementia patients could enhance safety and patient freedom. However, the technologies were found to have an increase of burden on staff in the care homes, since the technologies would give frequent, often unnecessary alarms, which is a design concern we will take into consideration.

Tasks and Content

Key Tasks:

Individuals experiencing memory or cognitive impairment have several important tasks that they need to complete throughout their daily lives. These tasks can range from simple everyday activities to tasks that can have serious consequences if forgotten. Some of the most important tasks include:

- Remembering to take medications at the correct time and dosage
- Remembering appointments, meals, and other scheduled activities
- Completing everyday routines such as eating, bathing, and getting ready
- Remembering where important personal belongings have been placed
- Navigating familiar environments without becoming lost or disoriented
- Recognizing when a task has already been completed to avoid accidentally repeating it
- Communicating with a caregiver when assistance is needed

Caregivers have a different but related set of tasks. They may need to remind the individual about upcoming activities, confirm that important tasks have been completed, locate the individual if they become lost, and determine when intervention is necessary.

Characteristics:

One of the most important characteristics of these tasks is that many are repetitive and time sensitive. Taking medication, for example, may need to happen every day at a particular time. Forgetting the task could be harmful but forgetting that the task was already completed could also cause the individual to accidentally repeat it.

The severity of memory impairment can also change how independently these tasks can be performed. One person may only need an occasional reminder, while another may require a caregiver to guide them through an entire routine. This means that the problem cannot be treated as through every person experiencing memory impairment has the same needs or abilities.

Another challenge is that remembering to use an aid can itself become a task. A person may have a calendar, reminder, or tracking device available but forget to check, charge, carry, or respond to it. This creates an important limitation for many existing solutions.

Task Environment:

Most of these tasks take place in everyday environments, particularly in the individual's home. The environment may contain physical reminders such as calendars, pill organizers, labels, clocks, and written notes alongside digital devices such as phones, tablets, smartwatches, or voice assistants.

The social environment is also important because caregivers, family members, healthcare professionals, and other people may participate in these tasks. Some individuals may live with a caregiver, while others may spend significant periods of time alone. Therefore, assistance may need to occur both when another person is physically present and when the individual is alone.

Tasks can also occur outside of controlled environments. An individual may need to remember an appointment while away from home or navigate back home after becoming disoriented. These situations introduce additional concerns because familiar physical reminders may no longer be available.

Structured Task Analysis: Hierarchical Task Analysis

The following Hierarchical Task Analysis (HTA) breaks the task of taking scheduled medication into smaller tasks and identifies where memory difficulties may interfere with completion.

- Goal 0: Take scheduled medication correctly
 1. Establish whether the scheduled medication needs to be taken.
 - a. Recognize that the scheduled time has arrived.
 - b. Check whether the scheduled dose has already been taken.
 2. Prepare the scheduled medication.
 - a. Locate the medication.
 - b. Identify the correct medication.
 - c. Identify the prescribed dose using the existing instructions.
 3. Take the scheduled medication.
 4. Record or communicate completion.
 - a. Use the existing tracking method or notify the caregiver that the dose was taken.
- Plan 0: Complete Task 1 first. If the doses are due and has not already been taken, complete Task 2-4 in order. If it has already been taken, stop. If the individual cannot establish whether it was taken or cannot identify the medication or prescribed dose, seek assistance before proceeding.
- Plan 1: Complete Tasks 1 in order.
- Plan 2: Complete Tasks 2 in order.

Memory impairment can cause a breakdown at almost any point in this process. The individual could forget that it is time to take the medication, forget where it is located, become uncertain

about the correct dosage, or forget whether they have already taken it. Looking at the task as a sequence rather than simply saying that the user needs to “remember their medication” helps identify exactly where memory-related difficulties interfere with the person’s ability to complete the task independently.

Research Methods and Evidence

To better understand the problem of memory and cognitive difficulties caused by aging or medical conditions, our research focused on understanding how memory and cognitive difficulties affect people's everyday lives and how technology has been used to provide support. We searched the ACM Digital Library for scholarly research related to older adults, cognitive impairment, memory support, dementia, and human-computer interaction. We also reviewed the information we could find from WHO which was statistics about the prevalence and effects of cognitive conditions.

Rather than searching only for "memory loss", we explored related topics such as cognitive impairment, aging, dementia, assistive technology, and reminders. This allowed us to examine a broader range of existing technologies and research approaches. The research also helped us understand how technology can be designed around the needs and abilities of people experiencing cognitive difficulties, while also showing some of the challenges involved in creating that technology that is accessible for older adults. We also used data from WHO to establish the scale of the issue. According to WHO, as of 2021, over 57 million people are living with dementia worldwide and every year there are almost 10 million new cases (World Health Organization). Dementia is known to affect memory and thinking, however, it also affects communication and a person's ability to perform everyday tasks.

We explored several areas of the technology space working with cognitive impairment. For example, Dong et al. Examined research trends surrounding the use of virtual reality for cognitive impairment. Their bibliometric analysis found growing research interest in using VR in areas such as cognitive impairment treatment, assessment, care, and rehabilitation. This helped us understand that there is already a broad research area with multiple possible approaches rather than an obvious technological solution. There was also research on people with mild cognitive impairment. Bouzida et al. Studied how people with mild cognitive impairment and their care partners experience frustration when interacting with technology and other systems. Their research found that participants experienced frustration across areas including communication, navigating changes in their abilities, and using technology. The researchers argue that these frustrations reveal accessibility barriers and opportunities for more inclusive technology design. This is particularly relevant to our problem because it shows that while the problem space has many solutions already provided, there are still issues that need fixing. We also looked at Hurstak et al.'s research on conversational agents for assessing mild cognitive impairment. The researchers tested four conversational assessment strategies with people with different levels of cognitive ability and compared results with the Mini-Mental state examination. One of the strategies demonstrated 92% sensitivity and 48% specificity compared with the Mini-Mental State Exam, while participants reported moderate satisfaction with the agent. This study provides evidence that conversational technology can potentially be used to interact with and assess

people experiencing cognitive difficulties, while also demonstrating that the effectiveness of such systems needs to be evaluated carefully.

Overall, our research explored the problem from multiple perspectives, the experiences and frustrations of people with cognitive impairment, existing technological approaches, and specific technologies such as conversational agents. This research helps demonstrate that researchers are investigating multiple technological approaches, but it also shows that usability, accessibility, and the experiences of people with cognitive impairment still have room for improvement. Combined with the health data from the WHO, this research provides evidence that memory and cognitive support is a significant problem area and gives us a foundation for identifying opportunities for technology that could support independence and everyday activities.

Synthesis and Design Implications

Our research has shown that one of the main issues with technology aimed at assisting those with cognitive/memory impairment is that they depend on the user remembering to use them. Tools like calendars, pill organizers, reminder apps, and gps trackers may be useful but won't be effective if the user forgets to use them. This creates an important challenge for when we go about designing technology for people who struggle to remember everyday tasks.

We also discovered that many of the tasks that people like this perform are repetitive and need to be completed on time. Taking Medication is an example because the user must remember when to take it, where it is, how much to take, and if they have already taken it. If they miss their meds or take too much, this could be catastrophic for their health. Because of this, our system should guide them through the action and confirm if it is completed.

Another important thing we discovered is that everyone with memory impairments will not need the same amount of assistance. Some people may only need reminders every once in a while, while others will need assistance from a caregiver. People suffering from these mental conditions that cause memory or cognitive impairment can become frustrated when using technology; it makes it worse if the technology is hard to use or confusing. This means the design must be simple and easy to understand. And not require users to jump through hoops just to complete a task.

Caregivers are also an important part of some solutions we could think of. Family members, friends, and professional caregivers may need to check if a task is completed, remind the person about something, or help locate them if they become lost or confused. Because of this, a future system should consider both the person with memory impairment and the caregiver. The user should still be able to have as much independence as possible while the caregiver can step in when needed.

The environment is also an important thing to consider when designing our system. Most of these everyday tasks happen at home where the person may have access to phones, tablets, smartwatches, voice assistants, or other devices. However, these people will not always be at home and may need help while they are somewhere else. Because of this, only using a phone app may not be the best option. We should consider different types of technology such as wearable devices, voice assistants, smart home devices, or a combination of these depending on what works best for the user.

Overall, the biggest opportunity we found is creating something that does not require the user to remember to use it in order for it to actually help them. Our design should be simple, require as little effort from the user as possible, and help them complete important everyday tasks. It should also allow caregivers to help when needed without taking away all of the user's independence. The main goal should be to make everyday life easier and safer without giving the user another thing that they have to remember.

Peer Feedback and Reflection

During our in-presentation, we presented three potential problem areas: Phishing and dark patterns, memory and cognitive impairment, and procrastination. Our peers' feedback highlighted strengths and concerns across all three areas and provided considerations for our selected focus on memory and cognitive impairment.

Key Themes in the Feedback

The main themes were the need to define a manageable problem, identify specific users, understand existing solutions, and consider whether people would consistently use a future system.

Some of the major considerations included:

- Impact: Peers recognized potential benefits across the ideas. One comment emphasized phishing prevention's broad usefulness, while several expressed interest in supporting motivation and reducing procrastination. Memory-related feedback raised a practical concern about helping older adults remember medication.
- Scope: One peer described the memory idea as too broad, and procrastination as insufficiently explored. Other comments called for a clearer approach to phishing prevention and a narrower target population for procrastination.
- Existing solutions: Repeated comments questioned how a procrastination project would differ from existing applications or another to-do list. A phishing comment also questioned whether the project would produce only a variation of an existing solution.
- Target users: Peers encouraged us to distinguish between general and technically experienced users for phishing, clarify the age group for procrastination, and investigate dementia-related behaviors for the memory idea.
- Feasibility: Concerns included incorrectly flagging legitimate websites, breaking large tasks into reasonable smaller steps, sustaining motivation, and whether people with memory difficulties would remember to use an app or device.

Feedback on Each Problem Area

- Phishing and Dark Patterns
 - o Peers saw potential for broad usefulness. This idea received seven selections for the strongest semester-long project potential, the three areas. All respondents agreed or strongly agreed that it supported at least three viable solution directions.

- Concerns focused on defining how phishing would be addressed, avoiding false alarms and missed threats, and distinguishing the project from existing approaches. Suggestions included exploring an awareness platform, identifying the intended audience, and clarifying whether an approach would depend on community input, AI, or an algorithm. Most written feedback addresses phishing rather than dark patterns specifically.
- Memory and Cognitive Impairment
 - All respondents agreed or strongly agreed that this problem supported at least three viable solution directions. Four respondents selected it for the strongest project potential, and four selected it as the idea they were most interested in seeing developed.
 - The written feedback highlighted scope and usability concerns. One peer described the idea as too broad. Another recommended researching the cognitive science behind dementia and common behaviors. A particularly relevant comment questioned whether older adults would remember to use an app or device intended to help them remember medication.
 - These comments suggest that we need to define the user population more precisely and investigate how memory difficulties affect the use of existing aids.
- Procrastination
 - Procrastination received the most selections for personal interest, with six respondents choosing it. It also received six selections for the strongest project potential. Peers responded to the possibility of helping people start earlier, maintain motivation, and recognize progress.
 - However, several comments emphasized the number of existing solutions and warned against creating another to-do list. Other concerns involved generating reasonable task breakdowns, implementing that functionality within the project, and retaining users whose preferences for rewards differ. Suggestions included emphasizing progress, narrowing the target population, and exploring distraction restrictions. Feedback about gamification was mixed, with some peers recommending rewards and another caution against excessive gamification.

How Peer Feedback Influenced Our Choice

Our team's selected problem area is memory and cognitive impairment. The feedback provides support for investigating this area while identifying limitations we need to address.

The comment about users forgetting to use an app or device reinforces a concern already identified in our initial exploration: memory aids may still depend on the very ability they are intended to support. This gives us a specific question for further research into the experiences of individuals with memory impairment and their caregivers.

The recommendation to investigate dementia-related behaviors also helps clarify the research needed before making design decisions. At this stage, the feedback supports further investigation of users, tasks, and existing aids without establishing that any technology will be effective.

Points of Disagreement or Tension

The feedback did not identify memory as the clear favorite. Phishing received the most selections for project potential, while procrastination received the most selection for personal interest. Some respondents selected multiple ideas for project potential, so those totals exceed the number of respondents.

There was also a tension between perceived opportunity and practical difficulty. Respondents consistently recognized multiple possible directions for memory, but written comments questioned its breadth and whether intended users could reliably engage with technology.

Our chosen focus therefore requires us to take these concerns seriously. The feedback supports exploring memory-related difficulties, but it does not establish that the problem is already sufficiently narrow or that a technology-based approach will be appropriate.

Implications for Future Work

The feedback suggests that our team should:

- Define the specific type or level of memory impairment and the everyday tasks without our project's scope.
- Investigate dementia-related behaviors where relevant to the selected user group.
- Examine when and why individuals forget to check, carry, or use existing memory aids.
- Distinguish the needs and responsibilities of individuals with memory impairment from those of caregivers.
- Evaluate existing approaches before assuming that additional reminders would address the problem.
- Use interviews, observations, and scholarly research to test these assumptions and determine a feasible scope for subsequent milestones.

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